

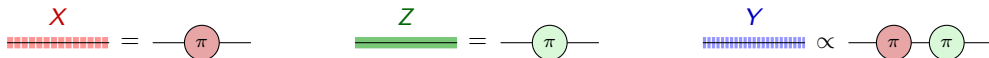
Pauli webs and semiwebs

A Pauli semiweb labels every wire of a ZX diagram with I , X , Y or Z . At each spider v the product w_v of the labels on its legs may shift its phase:

$$w_v |v(\theta_v)\rangle = \lambda_v |v(\theta_v + \delta_v)\rangle, \quad \delta_v \neq 0 \text{ is a defect } \bigcirc^*.$$

With no defect anywhere, the semiweb is a *Pauli web*: its labels leave every spider unchanged.

In **ZX terms** a label is a π spider on the wire:



A labelled wire inside the diagram carries its label twice, $PP = 1$, once for the spider at each end. Each spider then absorbs the π spiders on its legs. In a Pauli web every spider comes back as it was: the diagram is unchanged, and only the π spiders on its open legs are left over. A defect is a spider that comes back with another phase.

What a semiweb says. It relates the diagram D to the diagram D_δ in which the ringed phases are shifted.

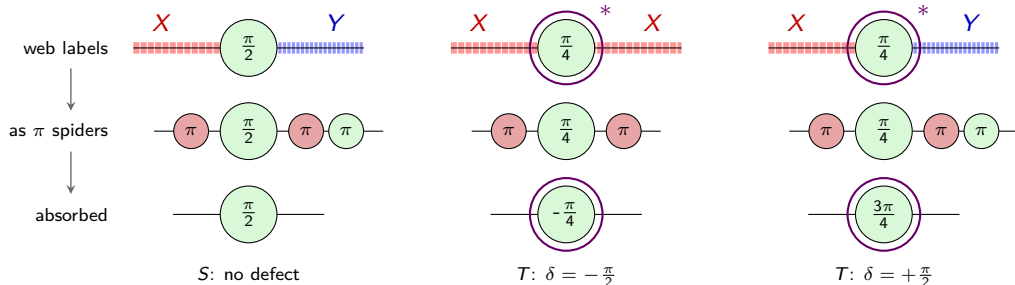
a closed web that reads the records c : $D_r = \lambda_0 (-1)^{c \cdot r} D_{r,\delta}$ for every record pattern r

an open web with a Pauli P on the output: $D = \lambda P D_\delta$

Kissinger and van de Wetering, arXiv:2603.09580. Notation of arXiv:2609.18922, Appendix I. Web edges: X Z Y .

A semiweb at one spider

The local identity $w_v|v(\theta_v)\rangle = \lambda_v|v(\theta_v + \delta_v)\rangle$ in pictures, for a Z spider with two legs. Read each column downwards: the labels of the web, the same labels as π spiders, and the spider after it has absorbed them.

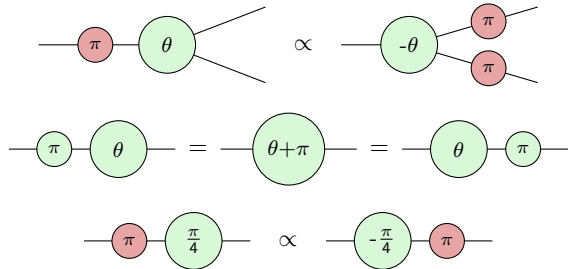


The S spider comes back as it was. With X on one leg the T spider does not, whether the other leg carries X or Y : it comes back as $T^\dagger$ or as $T S$, a defect of $\mp \frac{\pi}{2}$. The next two slides give the rules behind this.

The three examples are checked by a unit test. Web edges: ■■■■■ X ■■■■■ Z ■■■■■ Y . Defects: *.

Pushing a Pauli through a spider

The labels of a web are Paulis pushed through the diagram: Pauli propagation, drawn in ZX. At a Z spider of phase θ two rules of the ZX calculus do it.



Copy. X entering on one leg leaves on every other leg, and the phase is reversed: $\theta \mapsto -\theta$.

Fuse and unfuse. Z entering on one leg fuses with the spider, $\theta \mapsto \theta + \pi$. It can be unfused again onto any one other leg: Z passes through and the spider is as it was.

Copy at a T gate. X passes, and T becomes $T^\dagger$: $TX = e^{i\pi/4} X T^\dagger$.

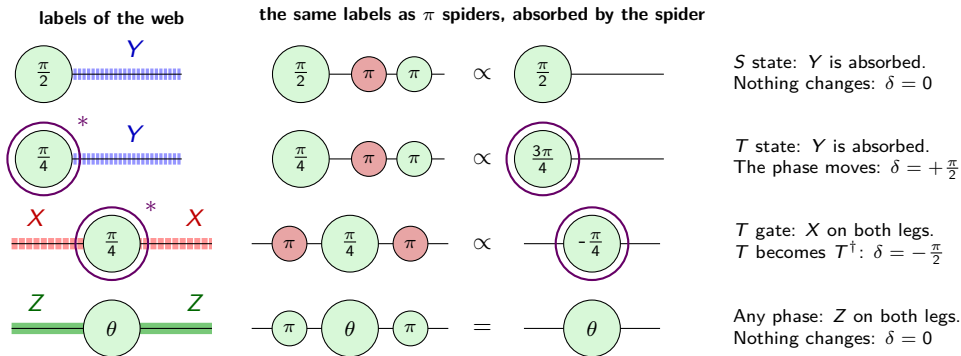
For an X spider exchange the two colours: there Z copies and X fuses.

Ordinary Pauli propagation writes the last line as $T^\dagger X T = \frac{X-Y}{\sqrt{2}}$: the gate stays and the Pauli splits into two terms. A semiweb keeps the single label X and moves the change into the spider. That change of phase is the defect.

The three rules are checked with matrices by a unit test. Pauli propagation and semiwebs: arXiv:2609.18922, Appendix I.

Where the defects come from

Read as propagation, the labels on the legs of a spider are Paulis going into it. A Z spider of phase θ absorbs them with $\delta = -2o\theta + \pi s$: $o = 1$ if its legs carry X or Y (copy: all of them or none), and s is the parity of the number of legs with Z or Y (fuse).

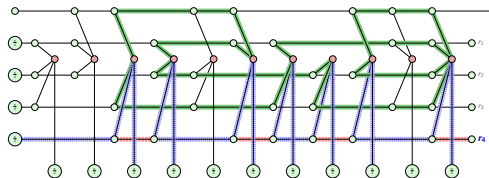
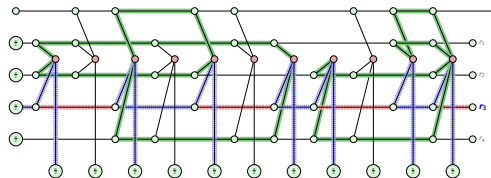
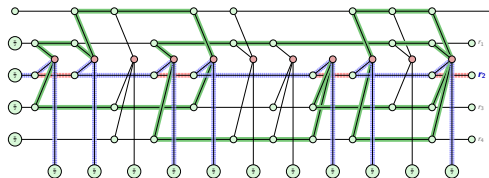
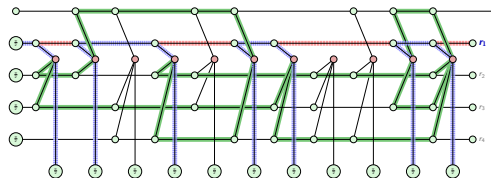


A label on both legs is a Pauli that passes through. At $\pm\frac{\pi}{2}$ the reversal costs π , and one green π repairs it. At $\pm\frac{\pi}{4}$ nothing repairs it: **an X or Y label at a T spider always leaves a defect of $\pm\frac{\pi}{2}$.**

The four examples are checked by a unit test. Web edges:  X  Z  Y. Defects: .

The four checks as Pauli webs

In the S -gate proxy every $\pi/4$ becomes $\pi/2$. Each X measurement $r_1, \dots, r_4$ is then a closed Pauli web.

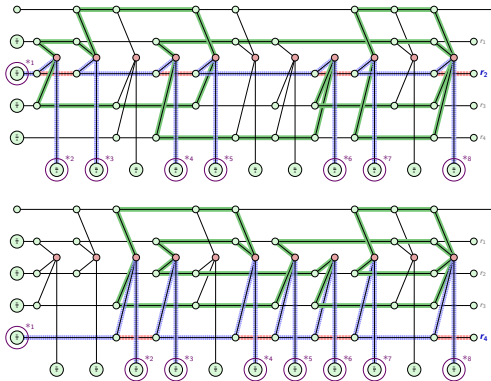
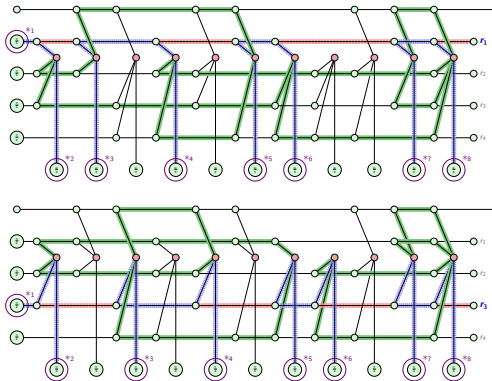


Each web is the measured X pushed back through the diagram until the S states absorb it as Y . The four generate every closed Pauli web of the proxy, and a Z fault on an input state flips the checks whose webs end on it.

Protocol: Bravyi and Kitaev, Phys. Rev. A 71, 022316 (2005). Circuit: Litinski, Quantum 3, 205 (2019). Webs computed from the diagram by `factory/run.py`. Web edges:  X  Z  Y.

With the actual T states: Pauli semiwebs

The same Pauli pushed back through the $\pi/4$ diagram arrives at 8 of the 15 T states as Y . A T state cannot absorb Y unchanged: each is left with a defect $\delta = +\pi/2$, ringed and numbered $*_1$ to $*_8$.



No check is a Pauli web here. Each holds only because its eight shifts cancel as a set ($D_\delta = D$). One $T \rightarrow T^\dagger$ makes the checks on its wires random.

Protocol: Bravyi and Kitaev, Phys. Rev. A 71, 022316 (2005). Circuit: Litinski, Quantum 3, 205 (2019). Semiwebs: Kissinger and van de Wetering, arXiv:2603.09580. Every identity is checked against the tensor of the diagram. Web edges: ■■■■■ X ■■■■■ Z ■■■■■ Y . Defects: ○ $*$.

Why each check still holds

Take the check r_i , the X measurement of wire i , and its semiweb with eight defects.

1. **The semiweb identity.** For a record pattern r , with c the records that the web reads,

$$D_r = \lambda_0 (-1)^{c \cdot r} D_{r,\delta}, \quad \lambda_0 = 1.$$

$D_{r,\delta}$ is the same diagram with the eight ringed phases moved from $\pi/4$ to $3\pi/4$.

2. **One shift is one extra S .** A T state that acts on the set s of wires multiplies $|x\rangle$ by $e^{i\frac{\pi}{4} p_s(x)}$, where $p_s(x)$ is the parity of the bits of x on s . Its shift multiplies $|x\rangle$ by $i^{p_s(x)}$ as well.
3. **The eight shifts together.** The eight T states are those that act on wire i . Mod 4 a parity is $p_s = \sum_{j \in s} x_j - 2 \sum_{j < k \in s} x_j x_k$, so

$$\sum_{s \ni i} p_s(x) \equiv \sum_j n_j x_j - 2 \sum_{j < k} n_{jk} x_j x_k \pmod{4},$$

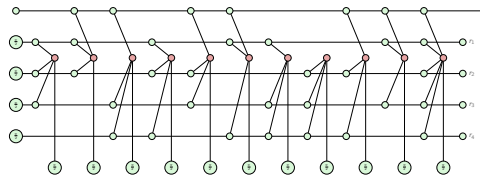
where n_j of the eight act on wire j and n_{jk} of them on both j and k . The factory's matrix gives $n_i = 8$, $n_j = 4$, and $n_{jk} = 4$ or 2 . Every term is $0 \pmod{4}$.

4. **So the shifts cancel.** $D_{r,\delta} = D_r$, hence $D_r = (-1)^{c \cdot r} D_r$. The diagram vanishes unless $c \cdot r = 0$: the check gives 0 with certainty.

Every step is checked in exact arithmetic by `factory/run.py`. The matrix has a row for each wire and a column for each T state: row weights 7, 8, 8, 8, 8, every two rows share 4 columns, every three share 2.

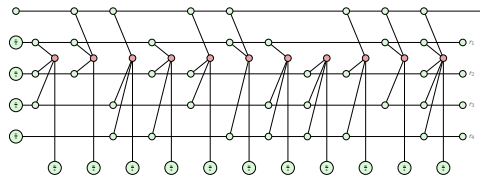
What the factory outputs

Contracted fully, with every X measurement at outcome 0, each diagram is one spider on the output wire.



$$\propto \text{red circle } \frac{\pi}{2} \text{ ---} = \text{green circle } \frac{\pi}{2} \text{ ---} \text{yellow square} \text{ ---} \propto \text{green circle } -\frac{\pi}{2} \text{ ---}$$

S -gate proxy: $|0\rangle - i|1\rangle$, the -1 eigenstate of Y .



$$\propto \text{green circle } -\frac{\pi}{4} \text{ ---}$$

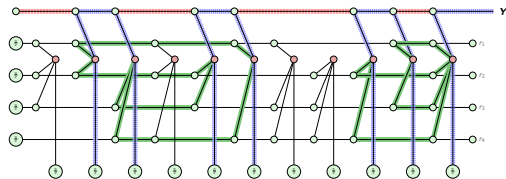
T states: $|0\rangle + e^{-i\pi/4}|1\rangle \propto T^\dagger|+\rangle$, the $+1$ eigenstate of $\frac{X-Y}{\sqrt{2}} = T^\dagger X T$.

The proxy output is a stabiliser state: an eigenstate of the Pauli Y . The output with T states is a magic state. It is not an eigenstate of X , Y or Z : its expectation values are $\langle X \rangle = \frac{1}{\sqrt{2}}$, $\langle Y \rangle = -\frac{1}{\sqrt{2}}$ and $\langle Z \rangle = 0$.

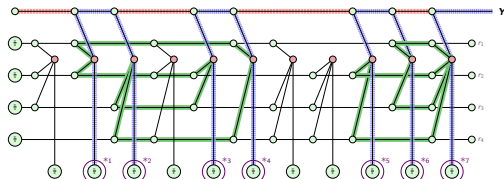
Protocol: Bravyi and Kitaev, Phys. Rev. A 71, 022316 (2005). Circuit: Litinski, Quantum 3, 205 (2019). $\propto$: equal up to a non-zero scalar. The tensor of each diagram is computed exactly and agrees with a direct five-qubit calculation.

The web of the output

An open web with a Pauli P on the output leg and defects δ says $D = \lambda P D_\delta$, with P acting on the output.



S-gate proxy: an open Pauli web, Y on the output



T states: 7 defects of $+\pi/2$, ringed and numbered $*_1$ to $*_7$

Proxy. No defect and $\lambda = -1$, so $D = -Y D$. The output is the -1 eigenstate of Y . A Pauli web can only end on the output with a Pauli of which the output is an eigenstate, and here that Pauli is Y .

T states. The same labels leave seven defects, on exactly the seven T states that act on the output wire. Now $D = \lambda Y D_\delta$ with $\lambda = e^{5i\pi/4}$, and the diagram on the right is a different one. The next two slides work out what this says about the output.

Protocol: Bravyi and Kitaev, Phys. Rev. A 71, 022316 (2005). Circuit: Litinski, Quantum 3, 205 (2019). Semiwebs: Kissinger and van de Wetering, arXiv:2603.09580. Web edges: ■■■■■ X ■■■■■ Z ■■■■■ Y . Defects: ○ $*$.

Step 1: the seven shifts are $S^\dagger$ on the output

The seven defects of the output web are the T states that act on wire 0, the output wire.

1. **Each shift is one extra S** , as for the checks: the shift of a T state on the wires s multiplies $|x\rangle$ by $i^{p_s(x)}$.
2. **The seven shifts together** multiply $|x\rangle$ by i to the power

$$\sum_{s \geq 0} p_s(x) \equiv \sum_j n_j x_j - 2 \sum_{j < k} n_{jk} x_j x_k \pmod{4}.$$

3. **Count with the matrix.** Now $n_0 = 7$, the weight of the output row. As before $n_j = 4$ for the other wires, and $n_{jk} = 4$ or 2 . Only one term survives: the power is $7x_0 \pmod{4}$.
4. **That is $S^\dagger$.** $i^{7x_0} = (-i)^{x_0}$: nothing happens when wire 0 is $|0\rangle$, and $|1\rangle$ gets a factor $-i$. The seven shifts are $S^\dagger = \text{diag}(1, -i)$ on the output wire and the identity on wires 1 to 4.
5. **Move it to the output.** All these operators are diagonal, and $S^\dagger$ on wire 0 commutes with the X measurements of the other wires. So

$$D_\delta = S^\dagger D.$$

For a measured wire the count is 8 and the shifts cancel. For the output it is 7, and one $S^\dagger$ is left.

Every step is checked in exact arithmetic by `factory/run.py`. It also compares the two tensors: the shifted diagram is $S^\dagger$ applied to the output.

Step 2: so the output is stabilised by $(X - Y)/\sqrt{2}$

1. **The semiweb identity** of the output web: $D = \lambda Y D_\delta$ with $\lambda = e^{5i\pi/4}$.
2. **Insert step 1**, $D_\delta = S^\dagger D$:

$$D = e^{5i\pi/4} Y S^\dagger D \quad \Longleftrightarrow \quad Y S^\dagger D = e^{3i\pi/4} D.$$

The output is an eigenstate of $Y S^\dagger$, with eigenvalue $e^{3i\pi/4}$.

3. **What is $Y S^\dagger$?**

$$Y S^\dagger = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ i & 0 \end{pmatrix} = e^{3i\pi/4} \begin{pmatrix} 0 & e^{i\pi/4} \\ e^{-i\pi/4} & 0 \end{pmatrix} = e^{3i\pi/4} \frac{X-Y}{\sqrt{2}}.$$

4. **The phases cancel:** $\frac{X-Y}{\sqrt{2}} D = D$. The output is the +1 eigenstate of $\frac{X-Y}{\sqrt{2}} = T^\dagger X T$, which is not a Pauli.
5. **That state is $T^\dagger|+\rangle \propto |0\rangle + e^{-i\pi/4}|1\rangle$:** , as found by contraction.

The proxy output is an eigenstate of the Pauli Y . The output with T states is an eigenstate of no Pauli: the operator that stabilises it is $\frac{X-Y}{\sqrt{2}}$.

Every step is checked in exact arithmetic by `factory/run.py`. A unit test repeats the matrix identities and both eigenvalue statements in floating point.